Docusign Envelope ID: D3EAC8FF-146A-4610-AFF4-26203B4BB02C

11 kV FROM HV SWITCHBOARD
F23-SW-601
2000-F23-SLD-EL-40001

F23-SR-601
FIM III BOOSTER STATION
HV/LV SWITCHROOM

F23-TR-602
TRANSFORMER ROOM

F23-TR-602
11/0.433kV
1600kVA
Dyn11
z=6%
ANAF

SWR EARTH BAR
Tx EARTH BAR
NOTE 3
4000/ 1A
2.5VA
5P20

F23-SW-602
FIM III BOOSTER STATION
LV MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
700/1A
2.5VA
(PART OF ELR PACKAGE)
NOTE 1
4000/5A
Cl1.0
(PART OF AGR-31C PACKAGE)
4000/ 1A
5VA
Cl1
TEST
PM
20A
63A
2A
63A
240VAC SPRING CHARGE SUPPLY
2A
ACB
AFD
4000A
24VDC CONTROL POWER SUPPLY
NOTE 7
4000/ 1A
5VA
Cl1.0
SD
NOTE 6
63A
100A

415V 3NE~ 50Hz 4000A 65kA for 1sec

INC

3F-17	4F-17	4F-14	4R-07	4R-10	3R-01	3R-04	3R-07	3R-10	4R-01	4R-04	3R-17						
250A 250A	250A 400A	63A 160A	32A 160A	32A 160A	20A 160A	20A 160A	20A 160A	20A 160A	20A 160A	20A 160A	400A 400A						
2A 20A																	
NOTE 14	NOTE 14	NOTE 14	NOTE 14	NOTE 14	NOTE 15	NOTE 15	NOTE 15	NOTE 15	NOTE 14	NOTE 14							

F23-DB-601
FIM III BOOSTER STATION
HV/LV SWITCHROOM
AND AREA L&SP DB
2000-F23-SCM-EL-40015
2000-F23-TLD-EL-40011

F23-DB-602
FIM III BOOSTER STATION
HV/LV SWITCHROOM
HVAC DB
2000-F23-SCM-EL-40016
2000-F23-TLD-EL-40012

F23-BC-601
FIM III BOOSTER STATION
HV SWITCHROOM
48V CHARGER
2000-F23-TLD-EL-40001

ISL
F23UPS601-ISL
FIM III BOOSTER STATION
240VAC UPS
MAINTENANCE
BYPASS SWITCH
2000-F23-SCM-EL-40017

10kW
F23-DB-603
FIM III BOOSTER STATION
LV SWITCHROOM
240VAC UPS DB
2000-F23-TLD-EL-40013

UPS
10kW
F23-UPS-601
FIM III BOOSTER STATION
LV SWITCHROOM
240VAC UPS
2000-F23-TLD-EL-40013

VS
9kW
F23PP651-VS
FIM III BOOSTER
PUMP 1
VARIABLE SPEED DRIVE
415V AUXILIARY SUPPLY
2000-F23-SCM-EL-40018

VS
9kW
F23PP652-VS
FIM III BOOSTER
PUMP 2
VARIABLE SPEED DRIVE
415V AUXILIARY SUPPLY
2000-F23-SCM-EL-40019

M 3~
1.5kW
F23-PW-651
FIM III
BOOSTER PUMP 1
SEAL WATER
PUMP 1
2000-F23-SCM-EL-40020

M 3~
1.5kW
F23-PW-652
FIM III
BOOSTER PUMP 1
SEAL WATER
PUMP 2
2000-F23-SCM-EL-40021

M 3~
1.5kW
F23-PW-653
FIM III
BOOSTER PUMP 2
SEAL WATER
PUMP 1
2000-F23-SCM-EL-40022

M 3~
1.5kW
F23-PW-654
FIM III
BOOSTER PUMP 2
SEAL WATER
PUMP 2
2000-F23-SCM-EL-40023

FL
1.0kW
F23-FL-600
FIM III
BOOSTER PUMPS
SEAL WATER
SELF CLEANING FILTER
2000-F23-SCM-EL-40024

SE
2.2kW
F23-SE-431
FIM III
BOOSTER STATION
SAFETY SHOWER
2000-F23-SCM-EL-40030

AHF
300A
F23-HF-601
FIM III BOOSTER STATION
LV MCC
ACTIVE HARMONIC FILTER
2000-F23-SCM-EL-40031

E-SHUT
F23SR601-LCS01
NOTE 12

E-SHUT
F23SR601-LCS02
NOTE 12

CONTINUED ON SHT 2 2000-F23-SLD-EL-40012

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F23-SW-602 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F23-BLK-IC-40001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/C WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.
13. CURRENT LIMITING FUSE SIZE TO BE CONFIRMED BY SWITCHROOM VENDOR.
14. COMPATIBLE 'TD' TOROID TO MATCH NHP ELR.
15. COMPATIBLE 'EL500S' TOROID TO MATCH AMPCONTROL ELV.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN

AS BUILT

REV	DATE	REVISION	DRN	CHK	TECH APP	APP
3	13.MAR.26	AS BUILT FOR FIM III TSF	[redacted]	[redacted]	MC	[redacted]
2	07.JUL.25	RE-APPROVED FOR CONSTRUCTION		[redacted]	KVN	
1	17.MAR.25	RE-APPROVED FOR CONSTRUCTION		[redacted]		
0	17.JAN.25	APPROVED FOR CONSTRUCTION				

REFERENCES	DWG No
F23-SW-602 LV MOTOR CONTROL CENTRE SINGLE LINE DIAGRAM SHT 2 OF 2	2000-F23-SLD-EL-40012
F23-SW-601 HV SWITCHBOARD SINGLE LINE DIAGRAM	2000-F23-SLD-EL-40001

NORTHERN STAR RESOURCES LIMITED

DRA PACIFIC (PTY) LIMITED

[redacted]	[redacted]	11.NOV.2024
CHECKED	J.Y.WEE	11.NOV.2024
DESIGNED	K[redacted]	11.NOV.2024
TECH APP	[redacted]	26.NOV.2024
[redacted]APP	T.HA[redacted]	17.MAR.2024
CLIENT APP		

PROJECT: KCGM
FIM TSF PROJECT

TITLE: FIM III BOOSTER STATION
F23-SW-602 LV MOTOR CONTROL CENTRE
SINGLE LINE DIAGRAM SHEET 1 OF 2

[redacted]1 | SCALE NTS | DRG No. 2000-F23-SLD-EL-40011 | REV 3